

Clinical Trial Complexity Assessment

OPAL / EFACCT Framework · 5 Weighted Dimensions · 34 Criteria · Score 0–3 per item

FRONT — SCORING CRITERIA

Score 0 — Low / None Score 1 — Low–Moderate Score 2 — Moderate–High Score 3 — High / Critical [See back for decision matrix & formula](#)

#	Criterion	Score 0 — Low / None	Score 1 — Low–Moderate	Score 2 — Moderate–High	Score 3 — High / Critical
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1	PROTOCOL DESIGN & STUDY COMPLEXITY	7 items · Weight 1.2× · Max raw 21 · Red flag >15
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1.1	Trial Phase	0 Phase IV / Observational	1 Phase III	2 Phase II	3 Phase I / First-in-Human
1.2	Study Design Complexity	0 Simple parallel / observational	1 Crossover / factorial	2 Adaptive randomization / dose-finding	3 Complex adaptive with multiple decision points
1.3	Number of Treatment Arms	0 Single / observational	1 Two arms	2 Three arms	3 Four or more arms
1.4	Eligibility Criteria Complexity	0 Broad (≤10 criteria)	1 Moderate (11–20 criteria)	2 Restrictive (21–30 criteria)	3 Highly restrictive (>30 or specialized testing required)
1.5	Biomarker Requirements	0 None required	1 Standard routine laboratory	2 Specialized send-out testing	3 Multiple specialized / novel assays
1.6	Primary Endpoint Complexity	0 Simple clinical or survival endpoint	1 Patient-reported outcome (PRO) / QoL	2 Composite or imaging endpoint	3 Novel endpoint / central adjudication required
1.7	Protocol Amendment History	0 Stable — no amendments anticipated	1 1–2 minor amendments	2 3–4 moderate amendments	3 >4 amendments or major design changes

2	OPERATIONAL EXECUTION & SITE BURDEN	8 items · Weight 1.5× · Max raw 24 · Red flag >17
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2.1	Number of Study Visits	0 ≤5 visits	1 6–10 visits	2 11–20 visits	3 >20 visits
2.2	Visit Duration	0 <1 hour — routine visit	1 1–2 hours — standard procedures	2 2–4 hours — multiple procedures	3 >4 hours — complex / inpatient
2.3	Assessment Procedures	0 Routine vitals and labs	1 Standard imaging or functional tests	2 Specialized procedures (PK, imaging)	3 Multiple specialized / invasive procedures
2.4	Data Collection Burden	0 Simple eCRF — minimal fields	1 Standard eCRF — moderate fields	2 Complex eCRF — extensive data points	3 Multiple systems / real-time / PRO support required
2.5	Monitoring Intensity	0 Remote or annual on-site monitoring	1 Every 6–12 months on-site	2 Every 2–6 months on-site	3 Monthly or 100% SDV (source data verification)
2.6	Investigational Product Handling	0 Oral drug — simple dispensing	1 Refrigerated storage or reconstitution required	2 IV infusion or complex preparation	3 Specialized storage / dedicated resources required
2.7	Third-Party Coordination	0 None required	1 Single vendor or CRO	2 Multiple vendors	3 Complex vendor network / decentralized components
2.8	Safety Reporting Requirements	0 Standard adverse event (AE) reporting	1 Enhanced AE monitoring	2 Real-time SAE / frequent reporting	3 Complex: DSMB / frequent safety labs / specialized reporting

3	REGULATORY & OVERSIGHT REQUIREMENTS	6 items · Weight 1.0× · Max raw 18 · Red flag >13
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3.1	IRB / Ethics Review	0 Standard review process	1 Expedited review — minor modifications	2 Full board review — vulnerable populations or sensitive	3 Multi-site or complex consent considerations
3.2	Regulatory Submissions	0 None required	1 Single submission (e.g., IND amendment)	2 Multiple or specialized submissions	3 Complex regulatory pathway or novel designation
3.3	Informed Consent Complexity	0 Standard single-signature consent	1 Lengthy consent / additional educational materials	2 Multi-stage consent or legally authorized representative (LAR)	3 Complex — multiple optional sub-studies or ongoing consent
3.4	Data Privacy Requirements	0 Standard HIPAA compliance	1 Enhanced security protocols	2 International data transfers or specialized requirements	3 Highly sensitive — exceptional security measures required
3.5	Oversight Committees	0 None required	1 Sponsor safety monitoring only	2 Independent DSMB or adjudication committee	3 Multiple committees / complex governance structure
3.6	Audit / Inspection Risk	0 Low risk — standard monitoring	1 Moderate risk — possible audit	2 High risk — likely audit	3 Very high risk — anticipated regulatory inspection

4	RESOURCE & WORKFORCE CAPACITY	7 items · Weight 1.3× · Max raw 21 · Red flag >15
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4.1	Coordinator Time Required	0 <0.25 FTE	1 0.25–0.5 FTE	2 0.5–1.0 FTE	3 >1.0 FTE or multiple coordinators needed
4.2	Specialized Expertise Needed	0 Standard site capabilities sufficient	1 Brief staff training required	2 Specialized experienced staff needed	3 Highly specialized skills or certification required
4.3	Space & Equipment Requirements	0 Standard exam room	1 Dedicated research space / standard equipment	2 Specialized equipment or procedure space required	3 Significant infrastructure investment required
4.4	Pharmacy Resources	0 None or minimal pharmacy involvement	1 Standard pharmacy services	2 Moderate — compounding or specialized storage	3 Intensive / specialized pharmacy capabilities required
4.5	Laboratory Resources	0 Routine commercial laboratory	1 Standard sample processing and storage	2 Specialized laboratory (e.g., –80°C freezer required)	3 Complex protocols / significant storage investment
4.6	Budget Adequacy	0 Budget clearly exceeds projected costs	1 Budget adequate with reasonable contingency	2 Marginally adequate — limited contingency	3 Budget insufficient for safe conduct
4.7	Current Site Workload	0 Significant available capacity	1 Moderate available capacity	2 Limited capacity — near maximum	3 At or over current capacity

5	PATIENT POPULATION & RECRUITMENT	6 items · Weight 1.0× · Max raw 18 · Red flag >13
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5.1	Target Population Accessibility	0 Large, readily accessible population	1 Moderate — accessible with standard outreach	2 Small — requires active recruitment effort	3 Very limited / rare disease population
5.2	Enrollment Timeline	0 Relaxed timeline / low enrollment rate	1 Moderate timeline and rate	2 Aggressive timeline / high enrollment rate	3 Very aggressive — likely unachievable
5.3	Competing Studies	0 No competing studies	1 Minimal competition — distinct populations	2 Moderate competition for same population	3 Direct competition / trial competes with standard of care
5.4	Patient Burden	0 Minimal burden — convenient participation	1 Moderate burden — manageable	2 Substantial burden — may deter enrollment	3 Excessive burden — significantly impacts consent
5.5	Retention Challenges	0 Low dropout risk / short study duration	1 Moderate retention challenge	2 Retention challenges anticipated	3 High dropout risk — likely to compromise data
5.6	Recruitment Strategy Feasibility	0 Proven strategy / high confidence	1 Standard approach / moderate confidence	2 Novel or uncertain strategy	3 Unclear / likely ineffective strategy

Clinical Trial Complexity Assessment — Reference & Decision Guide

How to calculate final score · Go / No-Go decision matrix · Automatic override conditions

BACK — DECISION GUIDE

GO / NO-GO DECISION MATRIX

Overall Score	Dim 5 Raw ≤ 6 (Low recruitment risk)	Dim 5 Raw 7–12 (Moderate risk)	Dim 5 Raw 13–18 (High risk)
0–40 Low	GO	GO	CONDITIONAL GO
41–65 Medium	GO	CONDITIONAL GO	NO-GO
66–100 High	CONDITIONAL GO	NO-GO	NO-GO

GO — Accept with standard oversight · CONDITIONAL GO — Accept with mitigation plan and senior sign-off · NO-GO — Decline (or escalate to leadership only)

DIMENSION WEIGHTS & SCORING FORMULA

Dimension	Items	Weight	Max Wt.	Red Flag
1. Protocol Design & Study Complexity	7	1.2×	25.2	>15
2. Operational Execution & Site Burden	8	1.5×	36.0	>17
3. Regulatory & Oversight Requirements	6	1.0×	18.0	>13
4. Resource & Workforce Capacity	7	1.3×	27.3	>15
5. Patient Population & Recruitment	6	1.0×	18.0	>13
TOTAL	34	—	124.5	—

- Step 1 — Raw dim score = Σ item scores (0–3 each)
Step 2 — Weighted score = raw $\times$ dimension weight
Step 3 — Total weighted = Σ all 5 weighted scores
Step 4 — Final (0–100) = total weighted $\div$ 1.245

AUTOMATIC NO-GO OVERRIDE CONDITIONS

If ANY condition below is met, the decision is NO-GO regardless of the total score:

- Any single dimension's weighted score $\geq$ 90% of its maximum possible
- Three or more dimensions with raw scores at or above their red-flag threshold
- Item 2.1 (Study Visits) = 3 AND Item 2.3 (Procedures) = 3 simultaneously
- Item 4.6 (Budget Adequacy) = 3 — budget clearly insufficient for safe conduct
- Item 4.7 (Current Workload) = 3 — site at or over current capacity
- Item 5.1 (Population Access) = 3 AND Item 5.2 (Enrollment Timeline) = 3

HOW TO COMPLETE AN ASSESSMENT

- Score each item Rate every criterion 0–3 using the Front side descriptors.
- Check red flags Any dimension raw score above its threshold triggers a warning.
- Apply weights Each dim score is multiplied by its weight factor (1.0 $\times$ –1.5 $\times$).
- Normalize Sum all weighted scores $\div$ 1.245 $\rightarrow$ final score on 0–100 scale.
- Apply matrix Cross-reference final score band with Dim 5 raw band at right.
- Check overrides Any override condition = automatic NO-GO regardless of score.

FINAL SCORE INTERPRETATION & THRESHOLDS

0 – 40 LOW COMPLEXITY — GO Site has sufficient capacity and experience. Proceed with standard oversight and normal monitoring frequency.	41 – 65 MEDIUM COMPLEXITY — CONDITIONAL GO Proceed only after documenting risk mitigation plans. Enhanced monitoring, dedicated coordinator time, and senior PI sign-off required.	66 – 100 HIGH COMPLEXITY — NO-GO Decline participation. If a strategic exception is considered, requires VP/C-suite approval with a full risk mitigation plan.
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